HIGH-SPEED PCB DESIGN: Do It Right The First Time

High-speed PCB design consists of multiple steps. Design re-spin is not only cost-sensitive but also time-consuming. So, one should follow proper design rules early in the design cycle to have the PCB working at the very first attempt

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Printed circuit board (PCB) is an essential part of all electronic goods, starting from electronic toys, remote controls, and mobile phones to complex electronic circuits in telecom equipment, aeroplanes, spacecraft, and medical devices. With rapid advancements in the electronics industry and the miniaturisation of components, PCB design has gone through a rapid evolution in the last few decades.

From PCBs of a few layers and mostly through-hole components, PCBs nowadays have tens of layers, fine-pitch components, and a mixture of through-hole and surface mount components. Modern complex designs at times require flexible PCBs and embedded components.

PCBs that are designed for high-speed signals (more than 1Gbps) need special attention during design as signal quality (rise/fall time, crosstalk, attenuation, etc) plays a crucial role in reliable operation. High-speed PCB design consists of multiple steps. Design re-spin is not only cost-sensitive but also time-consuming.

High-speed PCB design flow

It is better to follow proper design rules early in the design cycle to have the PCB working at the very first attempt. Fig. 1 shows the major steps involved in a high-speed PCB design. Let's go through the major steps one by one.

Schematic design. The design starts with a schematic drawing. There are many open sources as well as licensed software available for schematic design. But while designing a complex PCB design, it is better to go for reputed licensed versions as they offer better integration with the layout, useful features for quicker designs, and better constraint management.

During schematic drawing, the focus should be on functionality of the circuit and the selection of components. During selection of the components, the specifications to look for are size, power handling capacity, footprint, stray inductances and capacitances associated with that component, operating temperature range, and moisture sensitivity level (MSL).

Although the systems

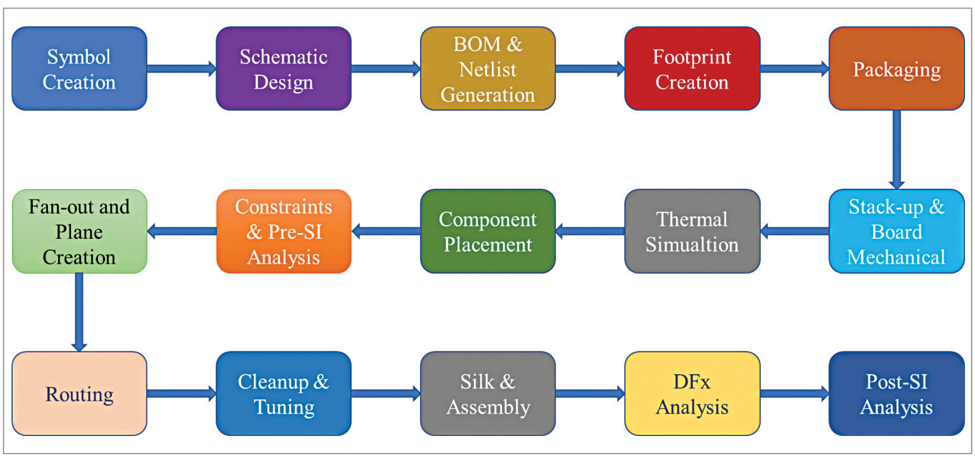

Fig. 1: Flow of high-speed PCB design